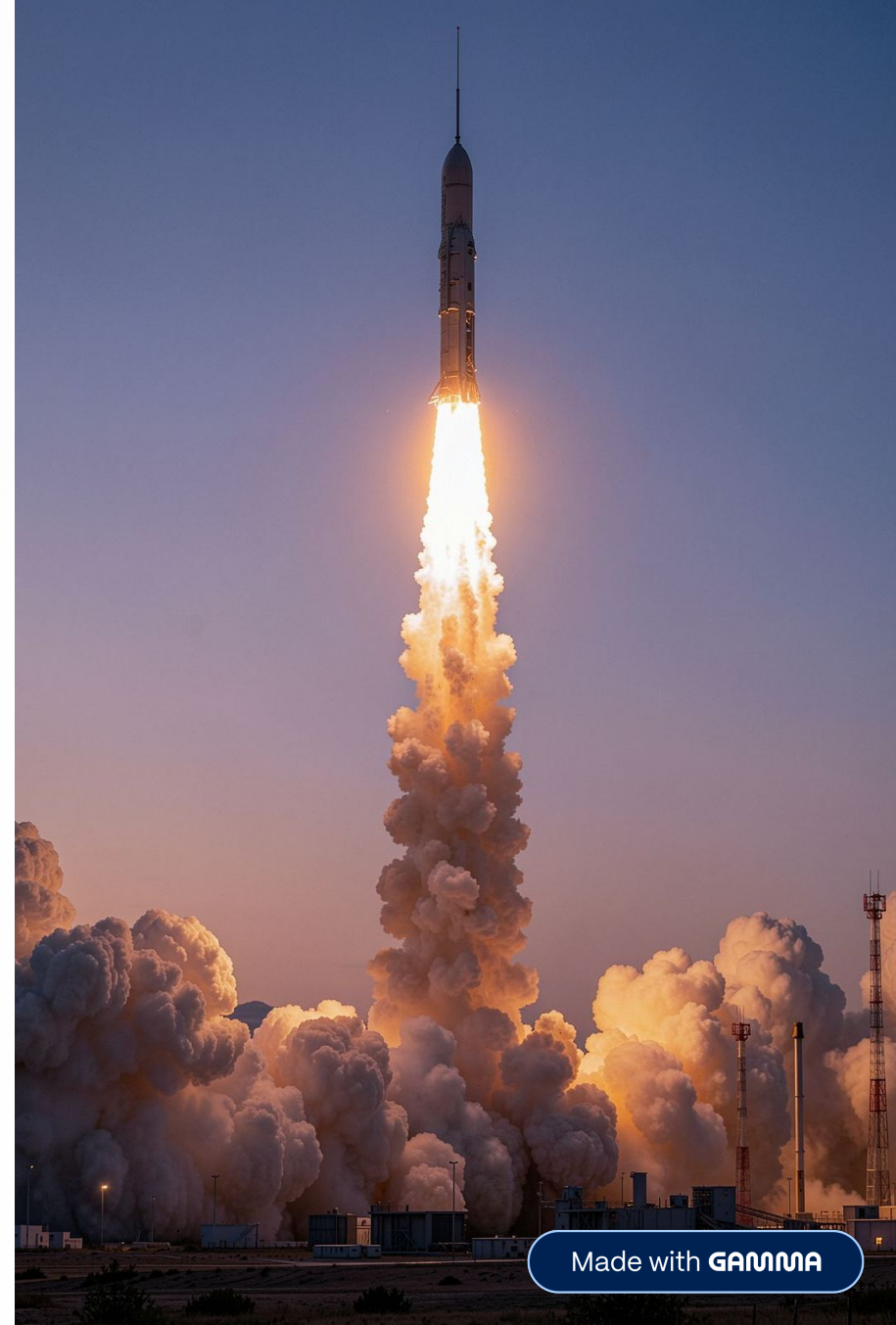


Space Mission Dataset - Complete EDA

An end-to-end analysis of 4,324 historical space missions, exploring mission activity, success rates, organizational performance, and mission costs. The analysis covers data understanding and preprocessing, exploratory analysis, machine learning-based price estimation, and interpretation of key trends to derive actionable insights.

[Start: Dataset Overview](#)[Open Report](#)



1. Dataset Summary & Schema

Rows: 4,324 | Original columns: 9. Key fields: Organisation, Location, Date, Detail, Rocket_Status, Price, Mission_Status. Two index-like columns (Unnamed: 0 and Unnamed: 0.1) were duplicates and provided no analytic value. Data types required normalization: numeric fields stored as text (Price) and dates stored as strings.



Rows & Columns

4324 rows, 9 source columns. Final dataset extends to 17 columns after feature extraction and imputation.



Problem Areas

Price stored as text (77.7% missing). Date strings require datetime conversion. Duplicate index columns removed.



Target Audience

Data analysts & data scientists performing reproducible EDA on mission-level datasets.

2. Data Preprocessing - Steps Taken

Drop redundant columns

Removed Unnamed: 0 and Unnamed: 0.1 to avoid duplicated index information and reduce noise.

Date normalization

Converted Date → datetime, extracted Year, Month, Decade for temporal analyses and aggregations.

Price cleaning

Stripped commas and non-numeric text; created Price numeric field and renamed to Price_Million_USD for clarity.

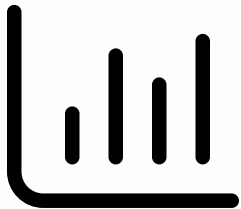
Feature extraction

Parsed Detail to produce Rocket_Name and Mission_Detail; standardized Organisation and Location strings.

Duplicate row checks were performed and removed where confirmed. Placeholder values were flagged and either standardized or marked for downstream review. All transformations were logged for reproducibility.

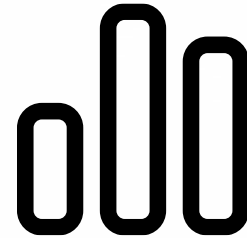
3. Organisation-level Activity & Price Distribution

Organisation counts: RVSN USSR leads mission counts by a wide margin, followed by Arianespace, General Dynamics, CASC, NASA, and VKS RF. Price distribution is right-skewed: most missions cluster around \$50–\$100M, a secondary mode near \$150–\$220M, and a small number of extreme missions up to multi-billion dollars.



Average & Median Prices

RVSN USSR shows an anomalously high mean and median ($\approx \$5,000\text{M}$ median) driven by limited, extreme records. NASA median $\approx \$450\text{M}$. Use medians and trimmed means when comparing organisations to reduce outlier influence.



Outliers

151 price outliers detected (149 NASA, 2 RVSN USSR). These were retained with flags because they likely reflect legitimately high-cost missions rather than entry errors.

4. Missing Values Strategy

Price had 3,360 missing entries (77.71%) - too many to drop without biasing results. We preserved the full dataset by imputing missing prices using two complementary approaches: an Extra Trees regression model trained on Organisation, Country, Year, and Month; and a Global Median fallback when model predictions were unreliable due to limited historical data for specific organisations.



Extra Trees Model

Used available price records ($n \approx 963$) to predict missing Price_Million_USD where organisation history supported modeling.



Global Median Fallback

Applied for organisations with insufficient price history to avoid extreme, unreliable estimates.



Provenance

Preserved original observed prices; created Estimated_Price_M and Price_Completed_M. Added Prediction_Method and Prediction_Reliability columns for transparency.

Imputation Results & Reliability

Outcome of imputation: 1,585 prices estimated by Extra Trees; 1,775 filled with Global Median fallback. Prediction_Reliability categories were assigned (Higher, Moderate, Low) based primarily on the amount of historical price data available for the organization. Over 2,200 estimates fall in Low reliability, ~340 Moderate, and ~760 Higher.



1,585 - Extra Trees

Model-based estimates where organisational history supported inference.



1,775 - Global Median

Conservative fallback to avoid extreme extrapolations for sparsely represented organisations.



Metadata

Each imputed row retains method and reliability to support sensitivity analyses and downstream filtering.

ML Model Evaluation

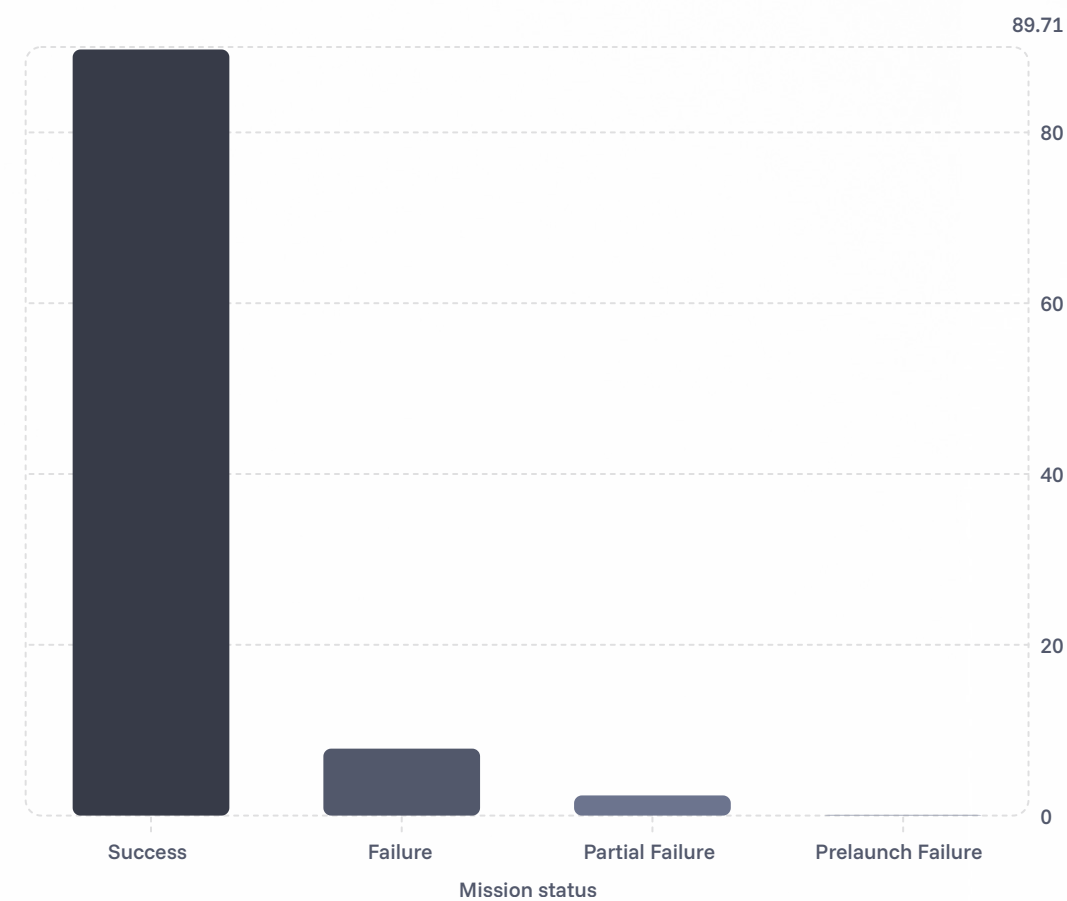
Model	MAE	RMSE	R ²
Linear Regression	2.45	18.85	0.9855
Random Forest	2.17	10.15	0.9958
Extra Trees	1.72	15.09	0.9907

Random Forest was evaluated as a baseline ensemble regression model using the same feature set. Although it achieved strong holdout performance, **Extra Trees produced a lower MAE (1.72M)** and was selected for the final estimation workflow. Extra Trees uses more randomized feature and split selection, which can improve variance reduction and generalization in this dataset.

Final Result Dataset

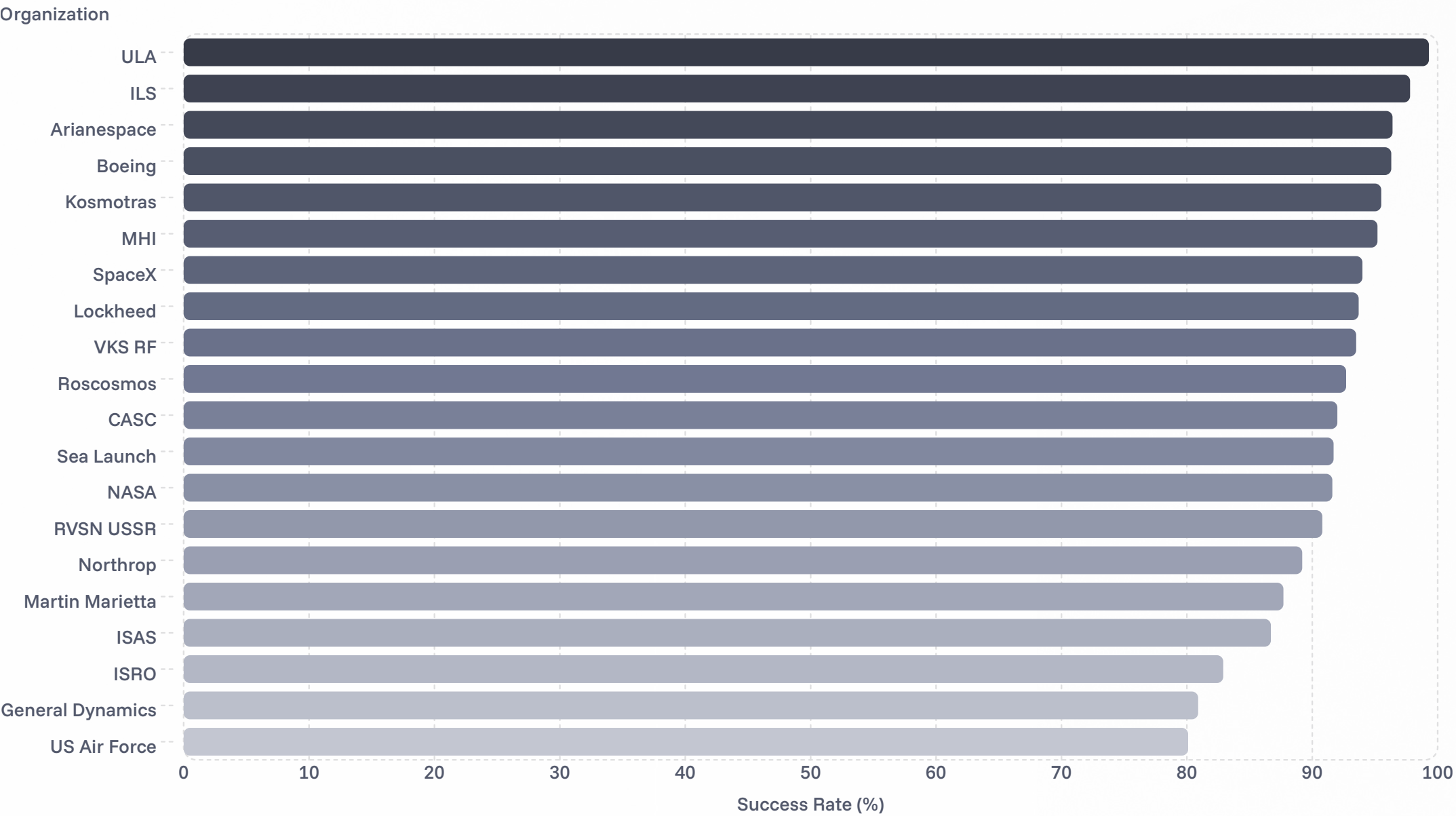
5. Key Exploratory Findings - Mission Outcomes and Status

Mission outcomes: Success ~89.7%, Failure ~7.8%, Partial Failure ~2.4%, Prelaunch Failure ~0.09%. The dataset is dominated by successful missions - interpret failure-related analyses with this class imbalance in mind.



Active rockets have a slightly higher mission success rate (**93.75%**) compared with retired rockets (**89.15%**). This suggests a modest association between active rocket status and higher mission success.

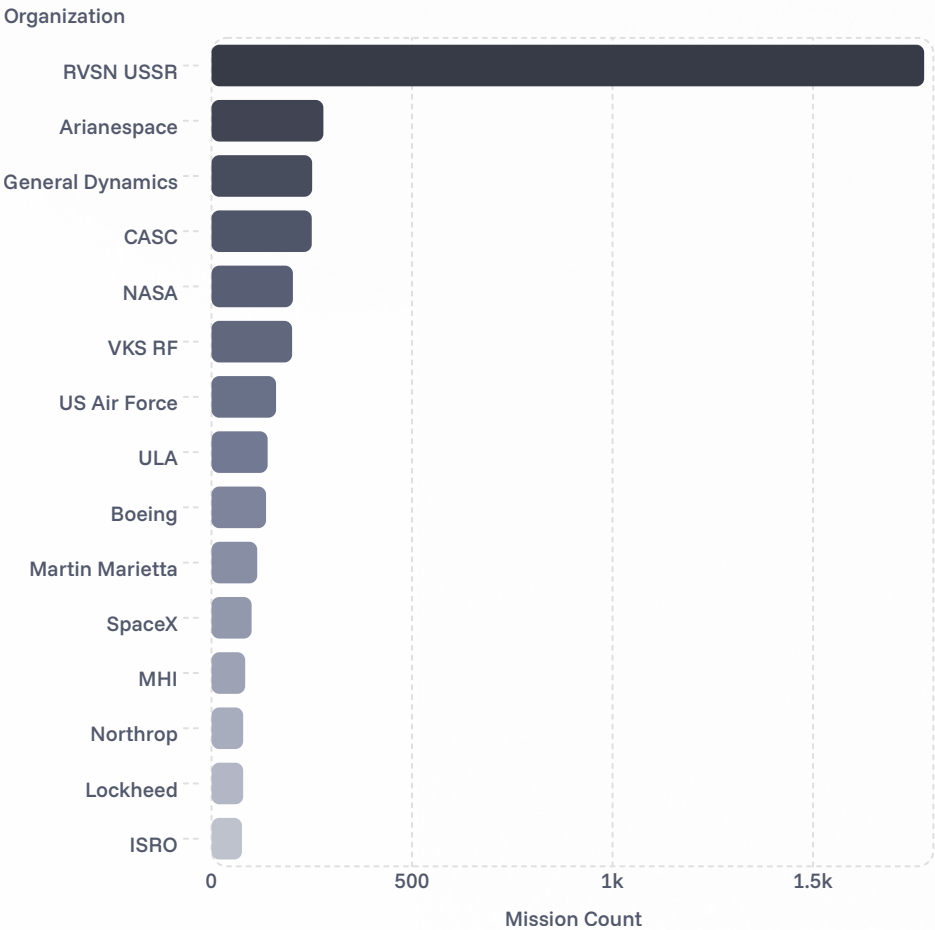
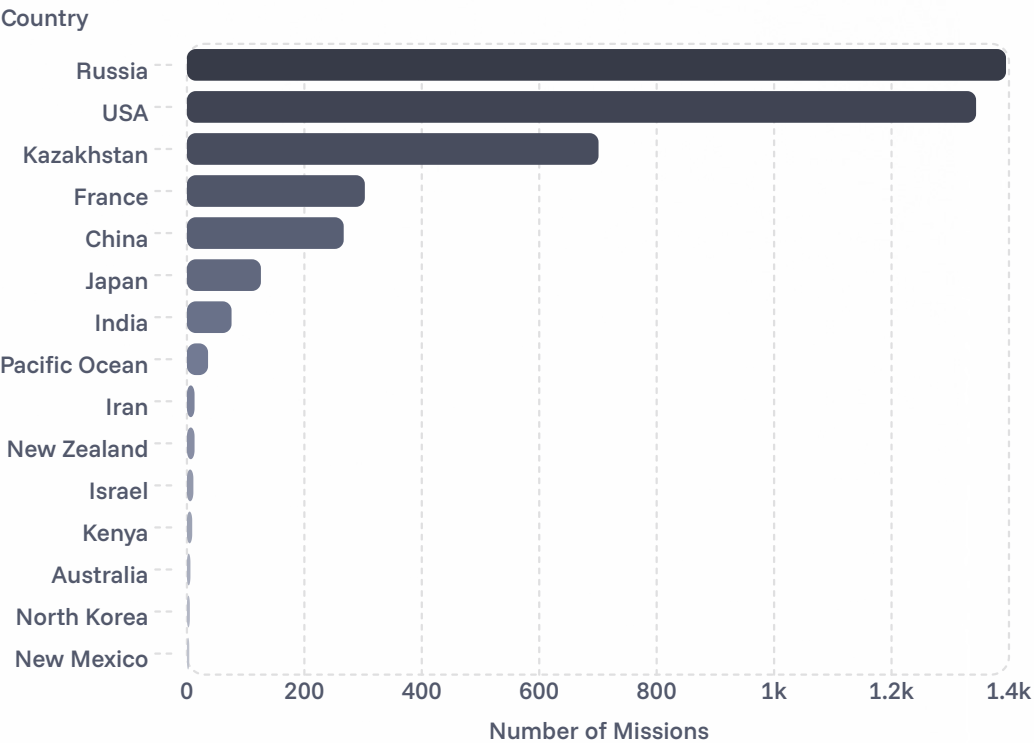
Organisation Performance Varies Across Major Launch Providers



Organizations with at least 20 missions show success rates ranging from **80.1% to 99.3%**. ULA and ILS record the highest success rates, while General Dynamics and the US Air Force are comparatively lower. Overall, most organizations maintain **above 90% mission success**, indicating generally high mission reliability.

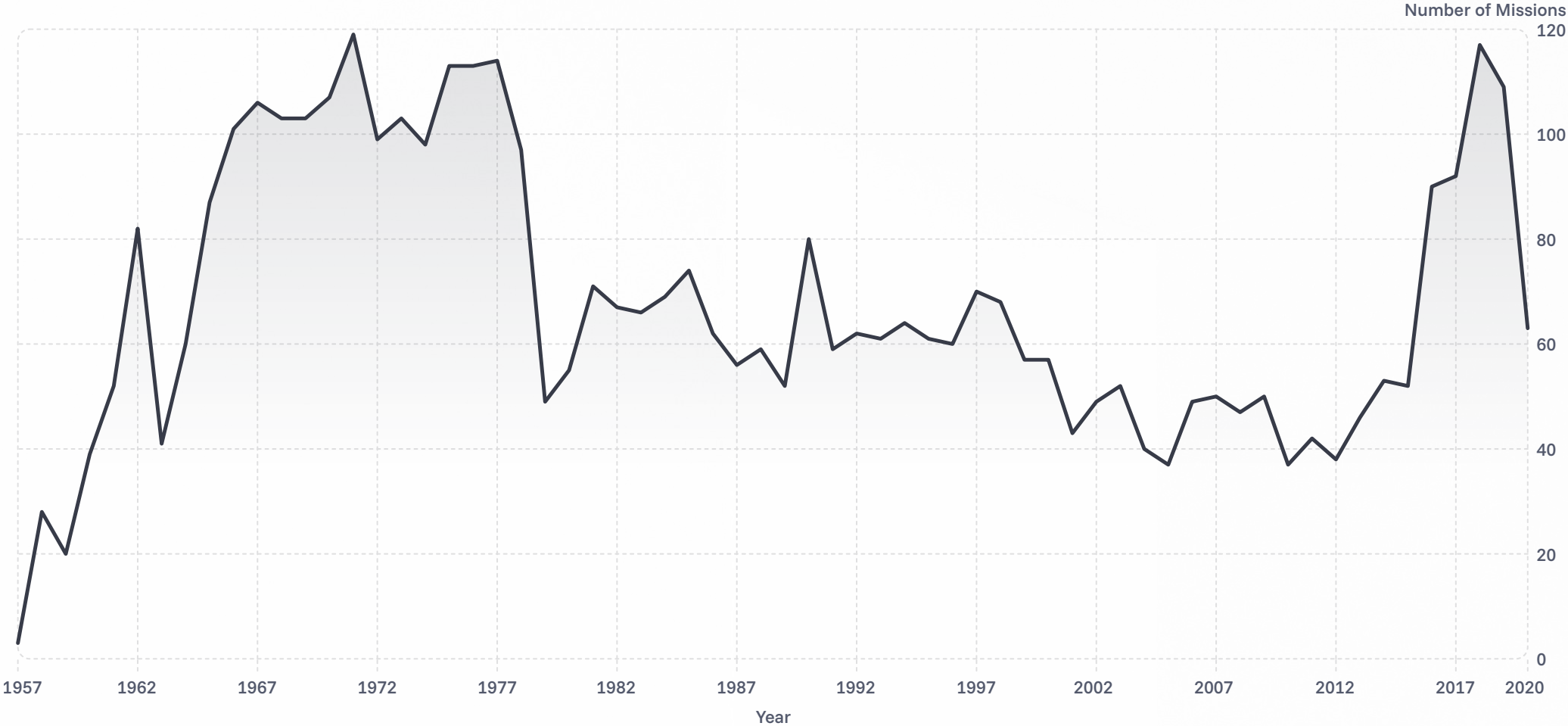
Mission Count Across Countries and Organizations

The chart shows that **Russia and the USA dominate space mission activity**, with around 1,400 and 1,340 missions respectively. **Kazakhstan ranks third**, followed by France and China, while the remaining countries have comparatively fewer missions, indicating a highly concentrated distribution of space activity.



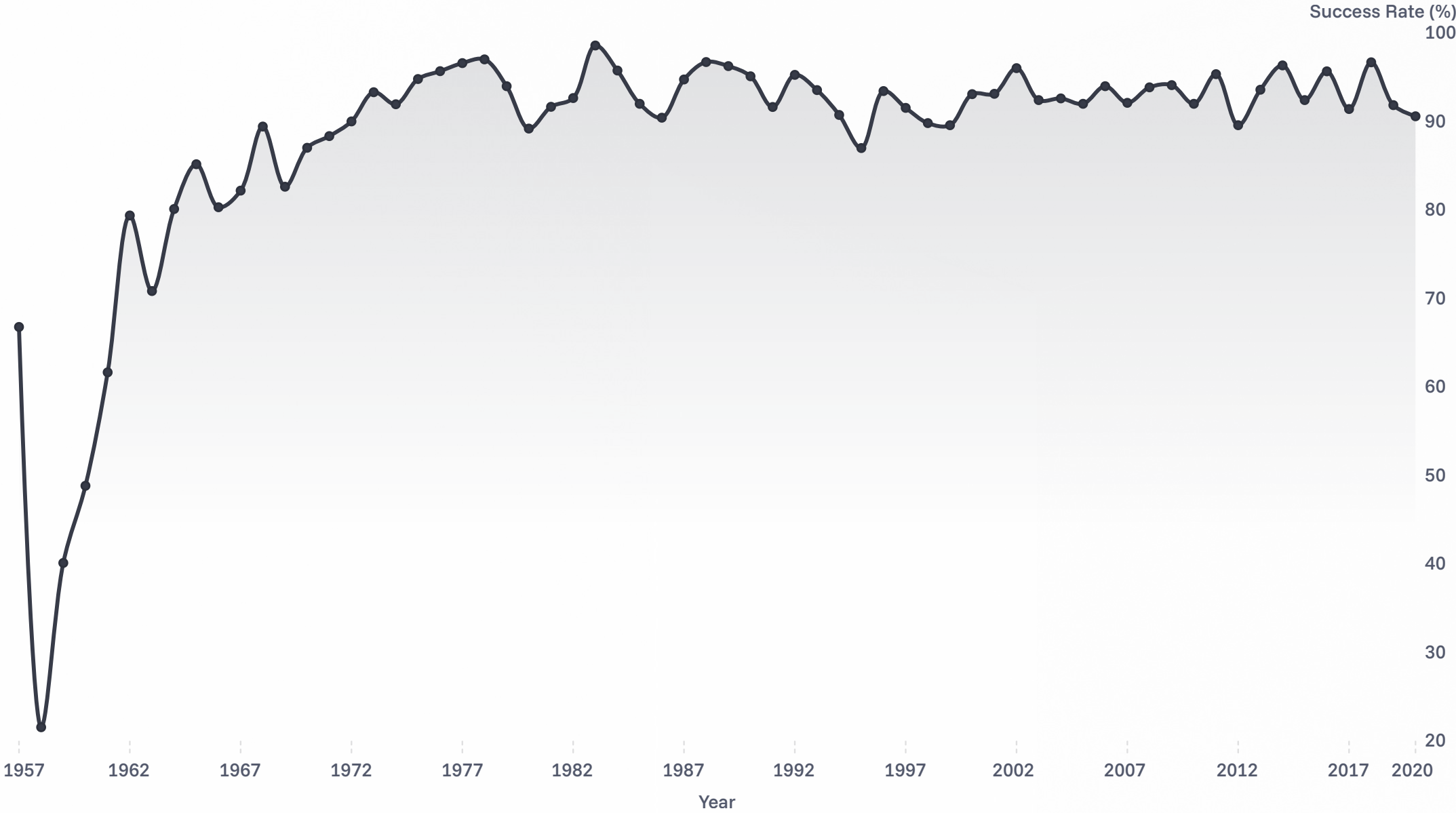
Analysts should stratify by country and decade to avoid conflating national program effects with temporal trends.

Mission Activity Increased Significantly Across the Space Era



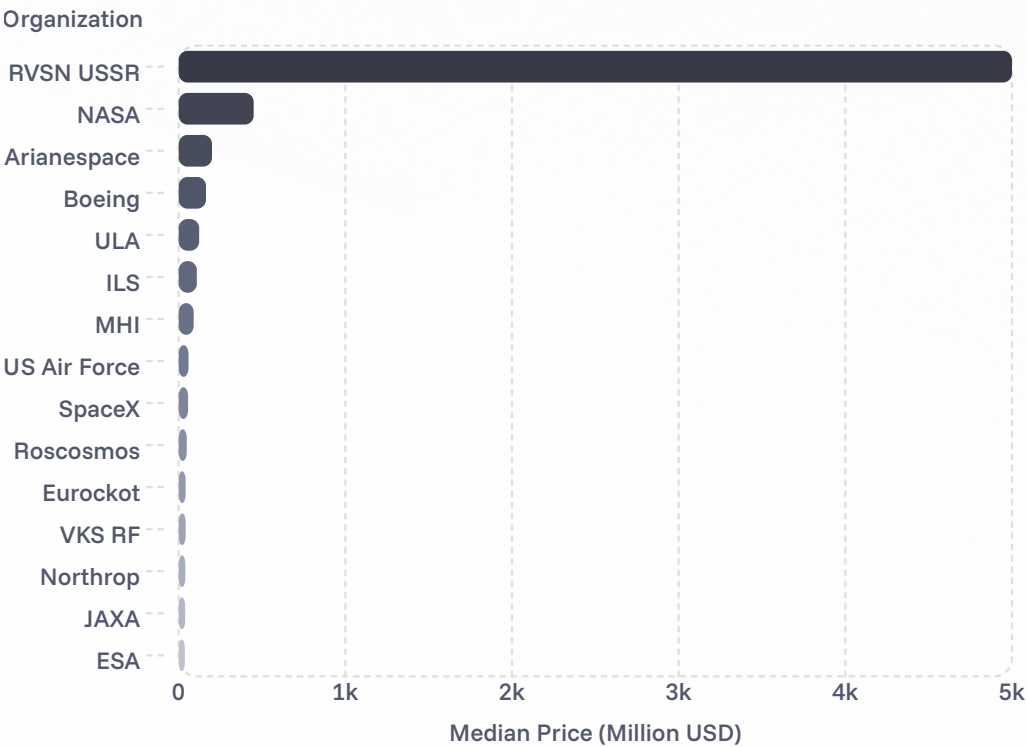
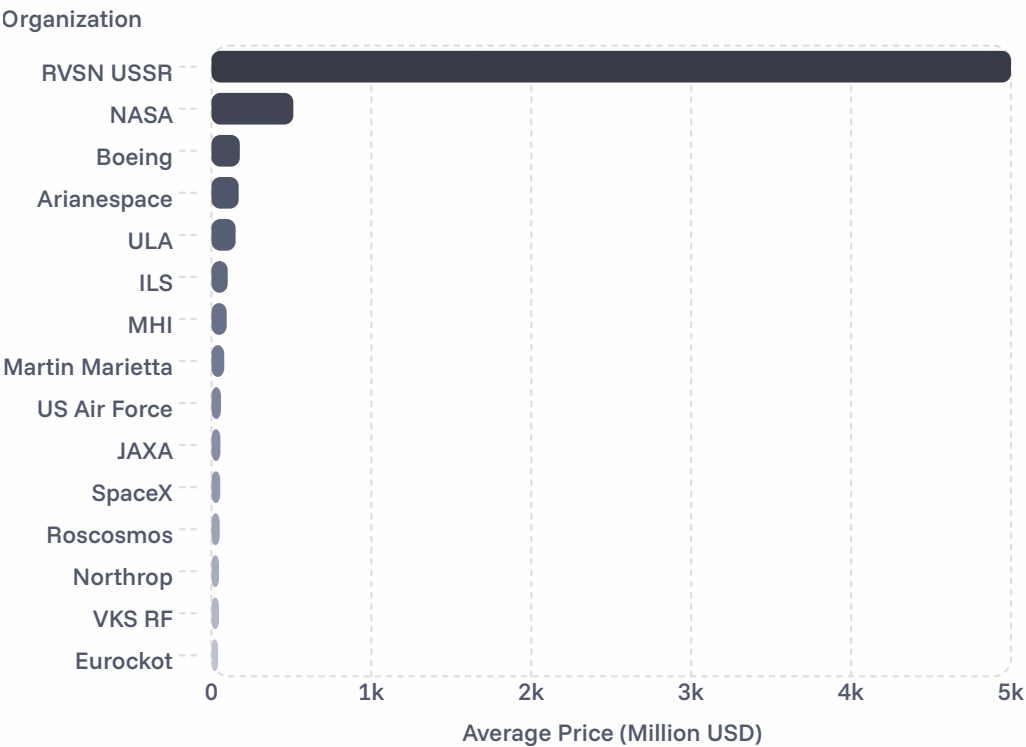
The number of space missions increased rapidly from the late **1950s to the 1970s**, reaching peaks of around 100+ missions per year. Activity declined during the **1980s–2010s**, followed by a sharp increase around **2017–2019**, before dropping again in 2020.

Consistent Success Rate Over Years



The mission success rate was relatively low during the **late 1950s and early 1960s**, but improved steadily through the 1970s. From the **1980s onward**, it **remained consistently high, generally above 85–90%**, reaching around 98% in some years.

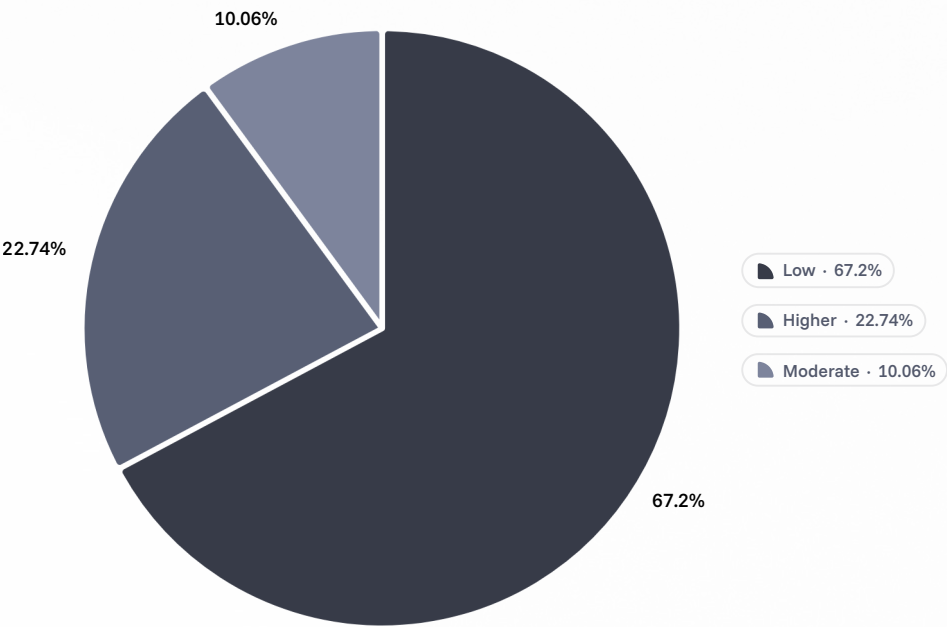
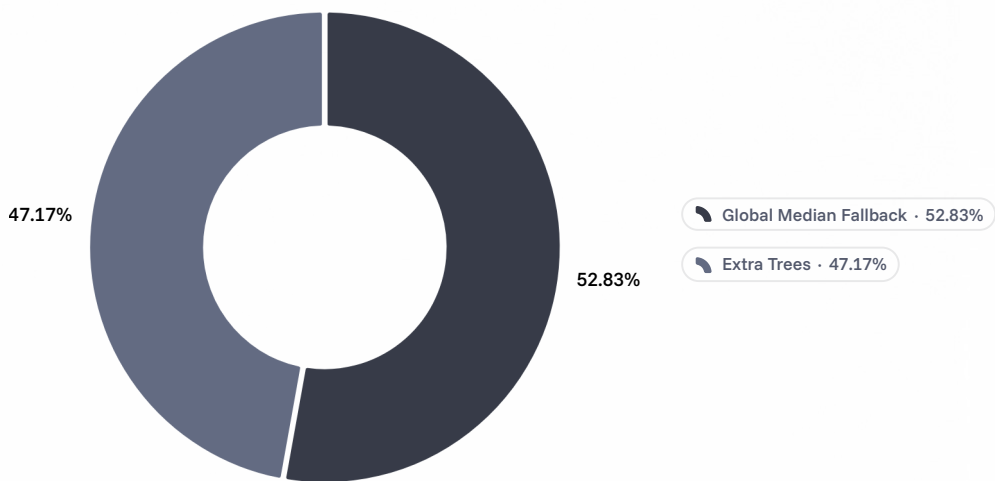
Average vs. Median Mission Cost by Organization



Average and median mission prices vary substantially across organizations, with RVSN USSR and NASA showing the highest values. Most other organizations have much lower typical mission costs, highlighting significant differences in mission pricing.

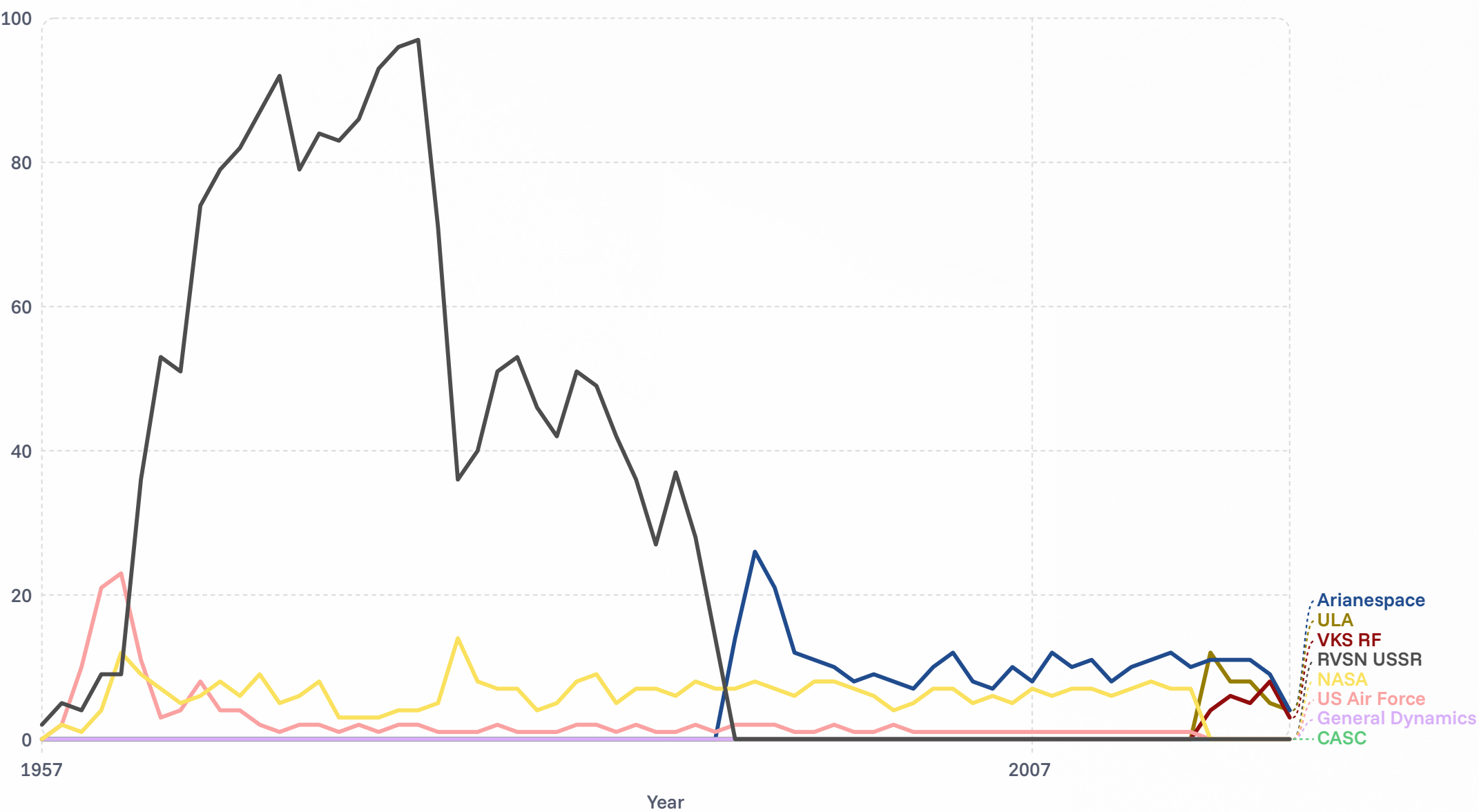
Model Performance - Price Estimation

The missing prices were estimated using two methods: **Global Median Fallback (1,775 missions)** and **Extra Trees (1,585 missions)**. Both model-based estimation and fallback estimation were required because historical price coverage varied substantially across organizations.



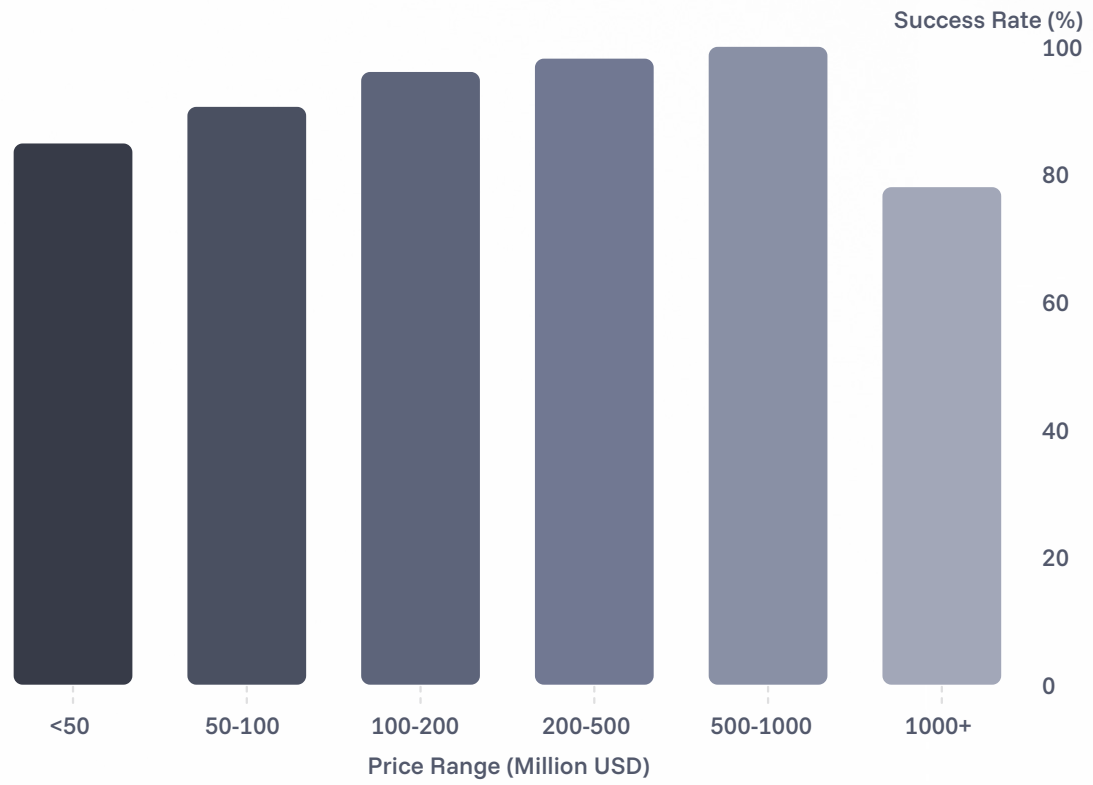
Most estimated prices fall into the **Low reliability** category, with over 2,200 predictions. Around **760 predictions have Higher reliability**, while approximately **340 are Moderate**. This indicates that a large portion of the estimated prices is based on limited historical pricing information and should therefore be interpreted with caution.

Mission Activity Trends by Organization (1957-2020)



The chart shows that **RVSN USSR dominated mission activity from the 1960s through the late 1970s**, reaching the highest annual mission counts. After the 1980s, its activity declined sharply, while organizations such as **NASA, Arianespace, CASC, VKS RF, and ULA** became more active. Overall, the pattern shows a **shift in space mission activity from early Soviet dominance toward greater participation by multiple organizations over time**.

Higher Recorded Price Ranges Show High Success Rates



The chart shows a **positive association between mission price and success rate**, with success rates generally increasing as price rises. However, this represents **correlation/association, not causation**-higher mission prices do not necessarily cause higher success rates, as other factors may influence mission outcomes.

6. Recommendations

1

Cost Benchmarking

Use historical mission costs to establish organization-level cost benchmarks.

2

Risk & Reliability Analysis

Use historical success rates to support mission planning and risk assessment.

3

Cost Estimation

Use ML-based estimates as a supporting tool when historical prices are unavailable.

4

Data Quality Improvement

Prioritize collecting actual mission-cost information because 77.7% of prices were missing.

5

Monitor Emerging Activity

Track changes in mission activity across organizations and countries to identify shifts in the space-launch landscape.

7. Limitations and Risks

- 77.7% of mission prices were missing, limiting direct cost analysis.
- Many estimated prices have **Low reliability**.
- RVSN USSR has very limited known price records and extreme values, which strongly affect price statistics.
- Historical mission data may not fully represent current launch economics.
- The price-success relationship shows **association, not causation**.

Presented by Veena M - [Contact here](#)